A B C A B C B C

A B A C B C

A B C

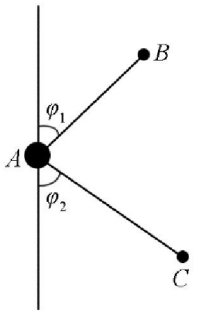
A B C A

AB AC φ_1 φ_2 $0<\varphi_1<\dfrac{\pi}{2}$ $0<\varphi_2<\dfrac{\pi}{2}$

AB $f_{AB}=k\left|AB\right|^a$ AC

$f_a=k\left|AC\right|^a$ $\left|AB\right|$ $\left|AC\right|$ A B A C

k a



15

(a)

(b)

(c)

$T_a=293\text{K}$ $T_1=273\text{K}$

$m=1.0\times10^{11}\text{ kg}$

1.